

CyberShield AI

AI-Powered Real-Time Security Operations Center (SOC)
Threat Intelligence Platform

Project Report

What Problem Does This Project Solve?

Modern organizations receive thousands or even millions of network packets every minute. A network administrator cannot manually inspect every packet. Because of this:

- Cyber attacks remain unnoticed.
- Threat detection is delayed.
- Network monitoring becomes difficult.
- Security teams receive alerts only after the damage is done.

CyberShield AI solves this by continuously monitoring network traffic, analyzing it using Machine Learning, detecting attacks in real time, and presenting the results through an interactive Security Operations Center (SOC) dashboard.

Real-World Example

Suppose a company's server is suddenly receiving thousands of ICMP packets. A human administrator may notice this only after the network becomes slow. CyberShield AI immediately:

- Detects abnormal traffic
- Identifies the attack type
- Calculates confidence
- Generates alerts
- Stores the incident
- Shows attack statistics on the dashboard

This reduces response time from minutes to seconds.

Main Objective

To develop an AI-based real-time network intrusion detection and monitoring system capable of detecting cyber attacks, classifying threats, generating alerts, and assisting administrators in responding quickly.

Target Users

- Security Operations Center (SOC)
- Network Administrators
- Ethical Hackers
- Cybersecurity Analysts
- Educational Institutions
- Small Enterprises

Technologies Used

Frontend

- React.js
- Vite

- React Router
- Axios
- Chart.js
- Leaflet Maps
- React Toastify
- CSS

Backend

- FastAPI
- Python
- SQLite
- JWT Authentication
- Passlib (Password Hashing)
- Scapy
- Psutil
- Pandas
- Joblib

Machine Learning

- Scikit-learn
- Random Forest Classifier
- Label Encoder

Deployment

Component	Platform
Frontend	Vercel
Backend	Render

Dataset Used

MSCAD — Multi-Step Cyber Attack Dataset

Contains multiple network attacks including:

- Normal Traffic
- Brute Force
- Port Scan
- HTTP DDoS
- ICMP Flood
- Web Crawling

Project Architecture

Internet Traffic



Packet Capture (Scapy)



Feature Extraction



Machine Learning Model (Random Forest)



Attack Prediction



■ Alert Engine

■ SQLite Database

■ Dashboard

■ Reports

Authentication Module

Purpose

Secure access to the SOC Dashboard.

Features

- Login
- JWT Token Generation
- Password Hashing
- Role Storage
- Protected Routes
- Admin Route

Database — Users Table

- ID
- Username
- Password (Hash)
- Role

Packet Capture Module

Aspect	Details
Uses	Scapy
Purpose	Captures network packets continuously
Collects	TCP, UDP, ICMP, Other packets, Total packets

These statistics are displayed on the dashboard.

Prediction Module

When prediction is requested:

Receives network features ↓ Loads trained ML model ↓ Predicts attack ↓ Returns Attack Name and Confidence Score

Example

Field	Value
Prediction	HTTP DDoS
Confidence	98%

Dashboard Module

Displays:

- Total Packets
- Active Threats
- CPU Usage
- Memory Usage
- Network Usage
- Packet Distribution
- Threat Pie Chart
- Live Predictions
- Alerts
- Recent Attacks

Live Traffic Module

Shows (updates continuously):

- TCP Packets
- UDP Packets
- ICMP Packets
- Other Packets

Analytics Module

Provides graphical analysis. Charts include:

- Threat Distribution
- Packet Distribution
- Prediction Statistics
- Historical Analysis

Alerts Module

Purpose: Immediately notify administrator.

Alert contains:

- Attack Type

- Time
- Confidence
- Severity

Incident Module

Stores every detected attack, including:

- Time
- Attack Type
- Confidence
- Status

Reports Module

Exports reports in the following formats:

- CSV
- Excel

Useful for: Audit, Documentation, Management Reports.

Activity Logs

Maintains system activities including:

- Login
- Predictions
- Alerts

Useful for auditing.

User Management (Admin Only)

- View Users
- Manage Users
- Role Control

Architecture Page

Explains the complete system workflow:

User ↓ Frontend ↓ Backend ↓ Machine Learning ↓ Database ↓ Dashboard

Settings Module

Contains application configuration, for example:

- Theme

● User Preferences

Database — SQLite

Table	Stores
Users	Username, Password, Role
Incidents	Attack, Time, Confidence, Status
Activity Logs	User actions

Machine Learning Pipeline

Dataset ↓ Cleaning ↓ Feature Selection ↓ Train/Test Split ↓ Random Forest Training
↓ Model Saving ↓ Prediction API ↓ Dashboard

API Endpoints

Category	Endpoint
Authentication	POST /login
Prediction	POST /predict
Dashboard	GET /dashboard
Traffic	GET /traffic
Attacks	GET /attacks
Reports	GET /download-csv GET /download-excel
System	GET /

Security Features

- JWT Authentication
- Password Hashing
- Protected Routes
- Admin Authorization
- CORS Protection

Deployment

Component	Hosted On
Frontend	Vercel
Backend	Render

Communication: REST APIs, JSON

Project Workflow

User Login ↓ JWT Authentication ↓ Dashboard Opens ↓ Packet Capture Starts ↓ Traffic Analysis ↓ ML Prediction ↓ Threat Detection ↓ Alert Generation ↓ Incident Storage ↓ Dashboard Update ↓ Report Generation

Advantages

- Real-time monitoring
- AI-powered detection
- Easy dashboard
- Fast response
- Low-cost solution
- Educational SOC platform
- Web-based
- Role-based authentication
- Interactive analytics

Current Limitations

- SQLite is suitable for small-scale deployments.
- Packet capture on cloud platforms is restricted because providers like Render do not allow raw socket access, so live capture is mainly intended for local deployment.
- The model detects only the attack classes it was trained on.
- The system is designed as a SOC monitoring and decision-support platform, not as a firewall or intrusion prevention system.

Future Scope

- Integrate SIEM platforms.
- Support PostgreSQL or MySQL for larger deployments.
- Add real-time email, SMS, or WhatsApp notifications.
- Implement automatic IP blocking through firewall integration.
- Introduce anomaly detection for unknown attacks.
- Add threat intelligence feeds and geolocation.
- Build multi-user dashboards with detailed audit trails.